



SAI VIDYA INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

SUBJECT: BASIC ELECTRONICS & COMMUNICATION

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MODULE - 4

Syllabus:

Analog and Digital Communication – Modern communication system scheme, Information source, and input transducer, Transmitter, Channel or Medium – Hardwired and Soft wired, Noise, Receiver, Multiplexing, Types of communication systems.

Types of modulation (only concepts) – AM, FM, Phase Modulation, Pulse Modulation, PAM, PWM, PPM, PCM.

Concept of Radio wave propagation (Ground, space, sky) Concepts of Sampling theorem, Nyquist rate, Digital Modulation Schemes– ASK, FSK, PSK

Radio signal transmission Multiple access techniques, Multipath and fading, Error Management

Antenna, Types of antennas (only definition and antenna model, **exclude radiation patterns**).

MODERN COMMUNICATION SYSTEM SCHEME:

- Communication engineering deals with the techniques of transmitting information.
- Communication engineering means electrical communication, in which information is transmitted through electrical signals.
- Electrical communication is a process by which the information message is transmitted from one point to another, from one person to another, or from one place to another in the form of electrical signals, through some communication link.
- Basic communication system provides a link between the information source and its destination. The process of electrical communication involves sending, receiving, and processing information in electrical form.
- The information to be transmitted passes through a number of stages of the communication system prior it reaches its destination.
- Figure 4.1 shows a block schematic diagram of the most general form of basic communication system.

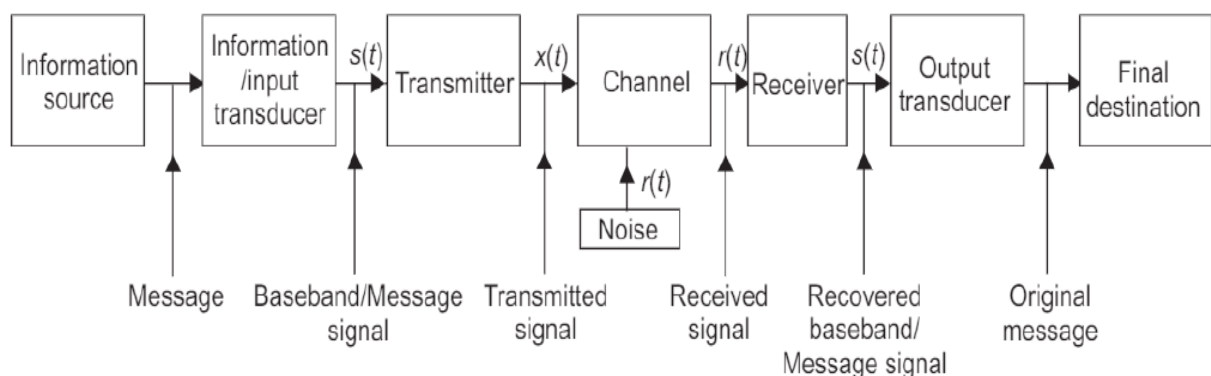


Fig 4.1 Schematic diagram of the most general form of basic communication system.

The main constituents of basic communication system are:

- (i) Information source and input transducer
- (ii) Transmitter
- (iii) Channel or medium
- (iv) Noise
- (v) Receiver
- (vi) Output transducer and final destination.

(i) Information source:

- A communication system transmits information from an information source to a destination and hence the first stage of a communication system is the information source.
- Ex: A sentence or paragraph spoken by a person is a message that contains some information. The

person, in this case, acts as information source. Few other familiar examples of messages are voice, live scenes, music, written text, and e-mail.

- A communication system transmits information in the form of electrical signal or signals.

(i) Input transducer:

- A transducer is a device that converts a non-electrical energy into its corresponding electrical energy called signal and vice versa, e.g., during a telephone conversation, the words spoken by a person are in the form of sound energy.
- An example of a transducer is a microphone. Microphone converts sound signals into the corresponding electrical signals.
- Similarly, a television (TV) picture tube converts electrical signals into its corresponding pictures. Some other examples of transducers are movie cameras, Video Cassette, Recorder (VCR) heads, tape recorder heads, and loudspeakers.
- The information produced by the information source is applied to the next stage, termed the information or input /transducer. This in turn, produces an electrical signal corresponding to the information as output. This electrical signal is called the baseband signal. It is also called a **message signal** $s(t)$.
- There are two types of signals. (a) analog signal, and (b) digital signal.

(a) Analog Signal

- An analog signal is a function of time, and has a continuous range of values. However, there is a definite function value of the analog signal at each point of time.
- A familiar example of analog signal or analog wave form is a pure sine wave form. A practical example of an analog signal is a voice signal. When a voice signal is converted to electrical for by a microphone, one gets a corresponding electrical analog signal.

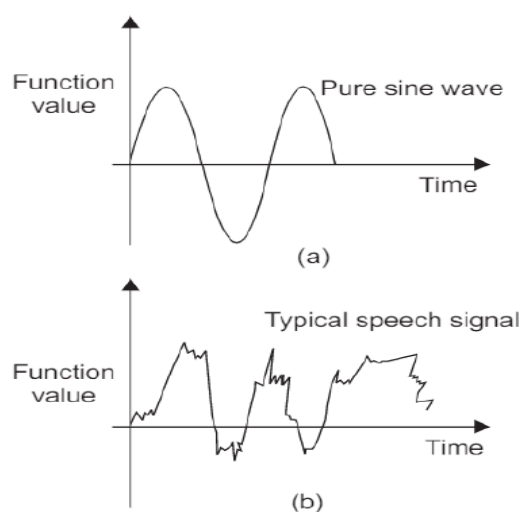


Fig 4.2: Analog signals (a) Pure sine wave (b) Typical speech signal

(b) Digital signal

- A digital signal does not have continuous function values on a time scale. It is discrete in nature, i.e., it has some values at discrete timings.
- A familiar example of a digital signal is the sound signal produced by drumbeats.

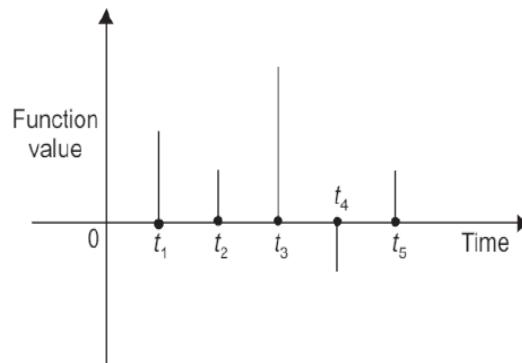


Fig 4.3: Digital signal

- Digital signals in their true sense correspond to a binary digital signal, where the discrete amplitude of the signal is coded into binary digits represented by '0' and '1'.
- The analog signal, which is continuous in time, is converted to discrete time, using a procedure called sampling. The continuous amplitude of the analog signal is converted to discrete amplitude using a process called quantization. Sampling and quantization are together termed as analog-to-digital conversion (ADC) and the circuitry that performs this operation is called an analog to-digital converter.

(ii) Transmitter:

The transmitter section processes the signal prior transmission. There are two following options for processing signals prior transmission:

- The baseband signal, which lies in the low frequency spectrum, is translated to a higher frequency spectrum.
- The baseband signal is transmitted without translating it to a higher frequency spectrum.
 - The baseband signal is converted into a corresponding series of sine waves of two different frequencies prior to transmission. Figure 4.4 illustrates this processing.
 - The carrier communication system is based on the principle of translating a low frequency baseband signal to higher frequency spectrum. This process is termed as **modulation**.
 - If the baseband signal is a digital signal, the carrier communication system is called a **digital communication system**. The digital modulation methods are employed for this.
 - If the baseband signal is an analog signal, the carrier communication system is called as an **analog communication system** and for processing the analog modulation techniques are used.

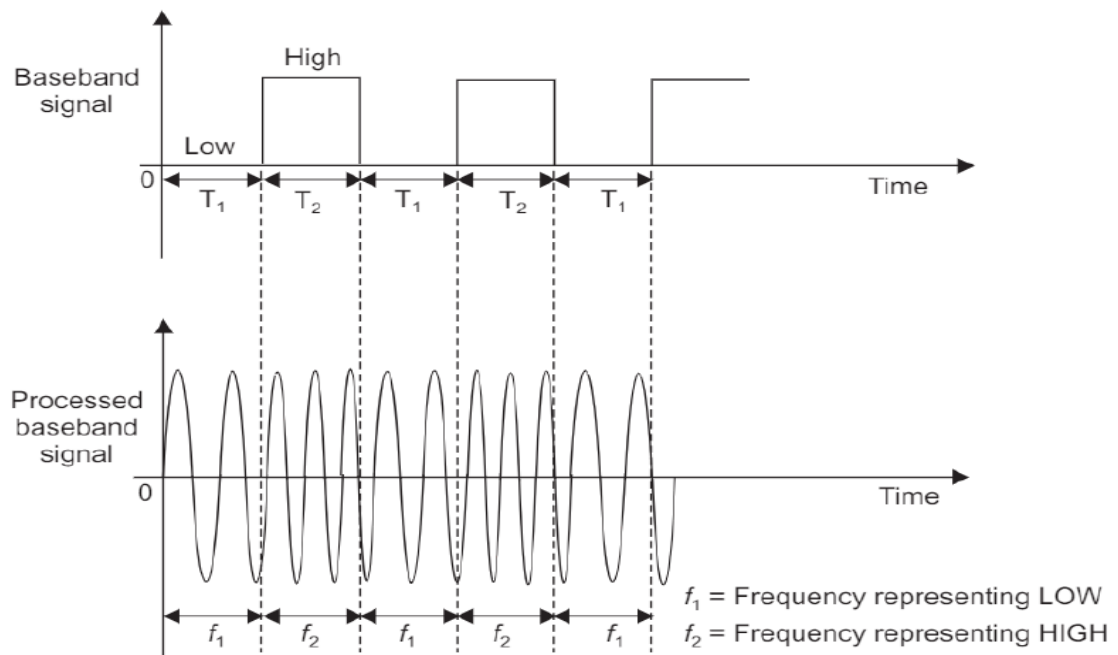


Fig: 4.4 The processing of a baseband signal

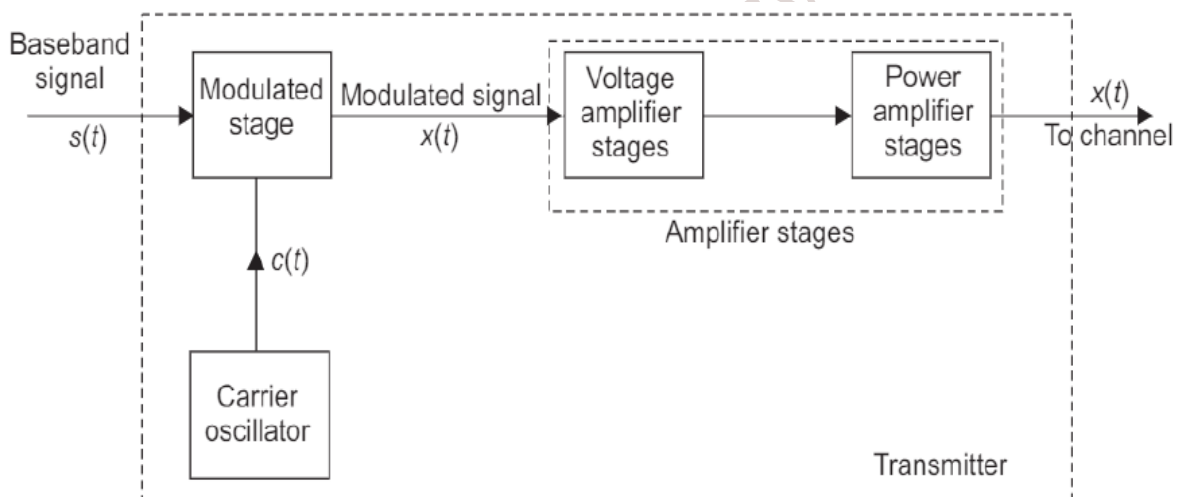


Fig. 4.5 Block diagram representing schematic of an analog transmitter section

- Figure 4.5 shows the baseband signal, $s(t)$ applied to the modulated stage. This stage translates the baseband signal from its low frequency spectrum to high frequency spectrum. This stage also receives another input called the *carrier signal*, $c(t)$, which is generated by a high frequency carrier oscillator.
- Modulation takes place at this stage with the baseband and the carrier signals as two inputs after modulation, the baseband signal is translated to a high frequency spectrum and the carrier signal is said to be modulated by the baseband signal.
- The output of the modulated stage is called the *modulated signal*, and is designated as $x(t)$. The voltage of the modulated signal is then amplified to drive the last stage of the transmitter, called the *power amplifier stage* (Fig. 4.5).

- This stage amplifies the power of the modulated signal and thus it carries enough power to reach the receiver stage of the communication system. Finally, the signal is passed to the transmission medium or channel.
- *Radio signals* are transmitted through *electromagnetic (em) waves*, also referred as radio waves, in a radio communication system.
- The radio waves have a wide frequency range starting from a few ten kilo Hertz (Hz) to several thousand Mega Hertz (MHz). This wide range of frequencies is referred as the *radio frequency* (RF) spectrum.

(iii) Channel or Medium:

- After the required processing, the transmitter section passes the signal to the transmission medium. The signal propagates through the transmission medium and is received at the other side by the receiver section. The transmission medium between the transmitter and the receiver is called a channel.
- **Channel** is a very important part of a communication system as its characteristics add many constraints to the design of the communication system, e.g., most of the noise is added to the signal during its transmission through the channel.
- Depending on the physical implementations, one can classify the channels in the following two groups:

Hardware Channels:

These channels are manmade structure which can be used as transmission medium. There are following three possible implementations of the hardware channels.

1. Transmission lines
2. Waveguides
3. Optical Fiber Cables (OFC)

- The examples of transmission lines are *Twisted-pair cables* used in landline telephony and coaxial cables used for cable TV transmission. However, transmission lines are not suitable for ultra-high frequency (UHF) transmission.
- To transmit signals at UHF range, *Waveguides* are employed as medium. Waveguides are hollow, circular, or rectangular metallic structures. The signals enter the waveguide, are reflected at the metallic walls, and propagate towards the other end of the waveguide.
- *Optical fiber cables* are highly sophisticated transmission media, in the form of extremely thin circular pipes. e.g., landline telephony and cable TV network.

Software Channels:

There are certain natural resources which can be used as the transmission medium for signals.

Such transmission media are called *software channels*.

- The possible natural resources that can be used as software channels are: *air or open space* and *sea water*.
- The most widely used software channel is air or open space. The signals are transmitted in the form of electromagnetic (em) waves, also called *radio waves*.
- Systems that use radio waves to transmit signals through open space are called radio communication systems, e.g., radio broad cast, television transmission, satellite communication, and cellular mobile communication.

(iv) Noise:

- In electronics and communication engineering, *noise* is defined as unwanted electrical energy of random and unpredictable nature present in the system due to any cause.
- Obviously, noise is an electrical disturbance, which does not contain any useful information. Thus, noise is a highly undesirable part of a communication system, and have to be minimized.

SNR and Noise Figure (F):

- One can define the SNR as the ratio of the signal power to the noise power at a point in the circuit. Now, if P_s is signal power and P_n is noise power, then SNR expressed as S/N , is given as

$$\left(\frac{S}{N}\right) = \left(\frac{P_s}{P_n}\right)$$

If $P_s = V_s^2 R$ and $P_n = V_n^2 R$, then

$$\frac{S}{N} = \frac{P_s}{P_n} = \frac{V_s^2 R}{V_n^2 R}$$

where V_s , is signal voltage and V_n , is noise voltage.

- In addition, it is assumed that both the signal and noise powers are dissipated in the same resistor R . Therefore, SNR can be expressed in terms of decibels (dB) as

$$\left(\frac{S}{N}\right) dB = 10 \log_{10} \left(\frac{V_s^2}{V_n^2}\right)$$

$$\left(\frac{S}{N}\right) dB = 20 \log_{10} \left(\frac{V_s}{V_n}\right)$$

- For example, if, at a particular point in a circuit, the signal and noise voltages are given as 3.5 mV and 0.75 mV, respectively, SNR in dB is calculated as:

$$\left(\frac{S}{N}\right) dB = 20 \log_{10} \left(\frac{3.5}{0.75}\right)$$

$$\left(\frac{S}{N}\right) dB = 20 \log_{10} (4.66)$$

$$\left(\frac{S}{N}\right) dB = 13.38 dB$$

Clearly, the SNR of the circuit at the point is 13.38 dB.

- The **Noise figure (F)** is the measure of the noise introduced by the circuit. It is defined as the ratio of the signal-to-noise power at the input of the circuit and the signal-to-noise power at the output of the circuit. Noise figure (r) can be expressed as

$$F = \frac{\frac{S}{N} \text{ Power at the input terminals of the circuit}}{\frac{S}{N} \text{ Power at the output terminals of the circuit}}$$

Receiver:

- The task of the receiver is to provide the original information to the user. This information is altered due to the processing at the transmitter side.
- The signal received by the receiver, thus does not contain information in its original form. The receiver system receives the transmitted signal and performs some processing on it to the original baseband signal.
- The function of the receiver section is to separate the noise from the received signal, and then recover the original baseband signal by performing some processing on it.
- The receiver performs an operation known as *demodulation*, which brings the baseband signal from the higher frequency spectrum to its original low-frequency spectrum. The demodulation process removes the high frequency carrier from the received signal and retrieves the original baseband.

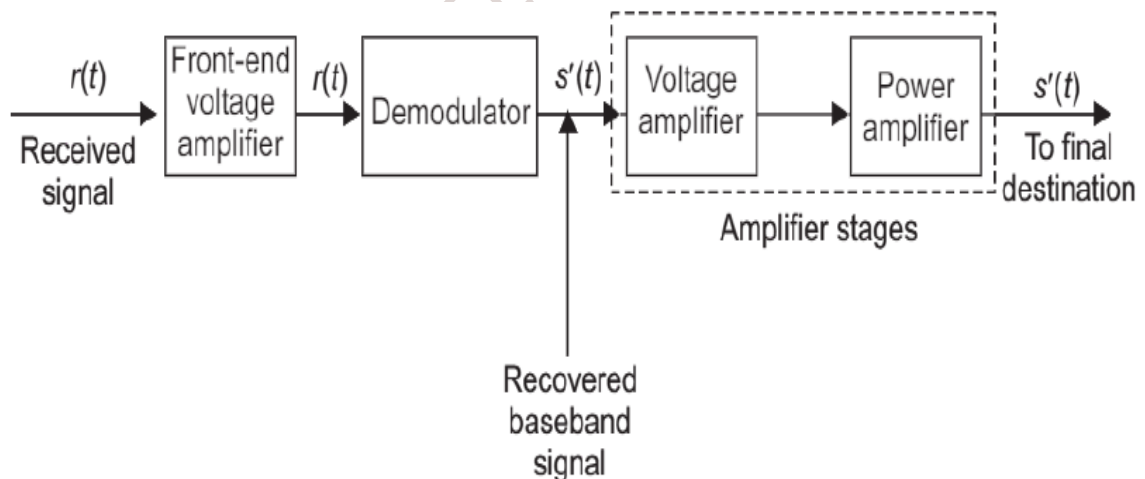


Fig: 4.6 Detailed block diagram of a typical receive section

- From Fig. 4.6 it is evident that the received signal, $r(t)$, is first amplified by the front-end voltage amplifier. This is done to strengthen the received signal, which is weak and to facilitate easy processing. Next, this signal is given to the demodulator, which in turn, demodulates the received signal to recover the original baseband signal. After recovering the original baseband signal, its voltage and power is amplified prior it to final destination block.

MULTIPLEXING

- This is a technique that is most widely used in nearly all types of communication systems, radio and line communication systems.
- Basically, multiplexing is a process which allows more than one signal to transmit through a single channel.
- The use of multiplexing also makes the communication system economical because more than one signal can be transmitted through a single channel.
- Multiplexing is possible in communication system only through modulation.
- To consider multiplexing, let us consider the following example. If many people speak loudly and simultaneously, then it becomes nearly impossible to understand their conversation because the overall result is noise. This noise is the result of mixing of all the speeches. The human ear is not capable of separating these intermingled speeches and therefore no intelligent words are communicated to brain. The same situation is now applied to the transmission of audio signals. These audio signals may come from, say ten different persons. While the speech frequency of different persons will be different, all the ten signals will lie in the same audio range of 20 Hz to 20 kHz.

TYPES OF COMMUNICATION SYSTEMS

One may categorize communication systems based on:

- The *physical infrastructure* pertains to the type of the channel used and the hardware design of the transmitting and receiving equipment.
- The *signal specifications* signify the nature and type of the transmitted signal

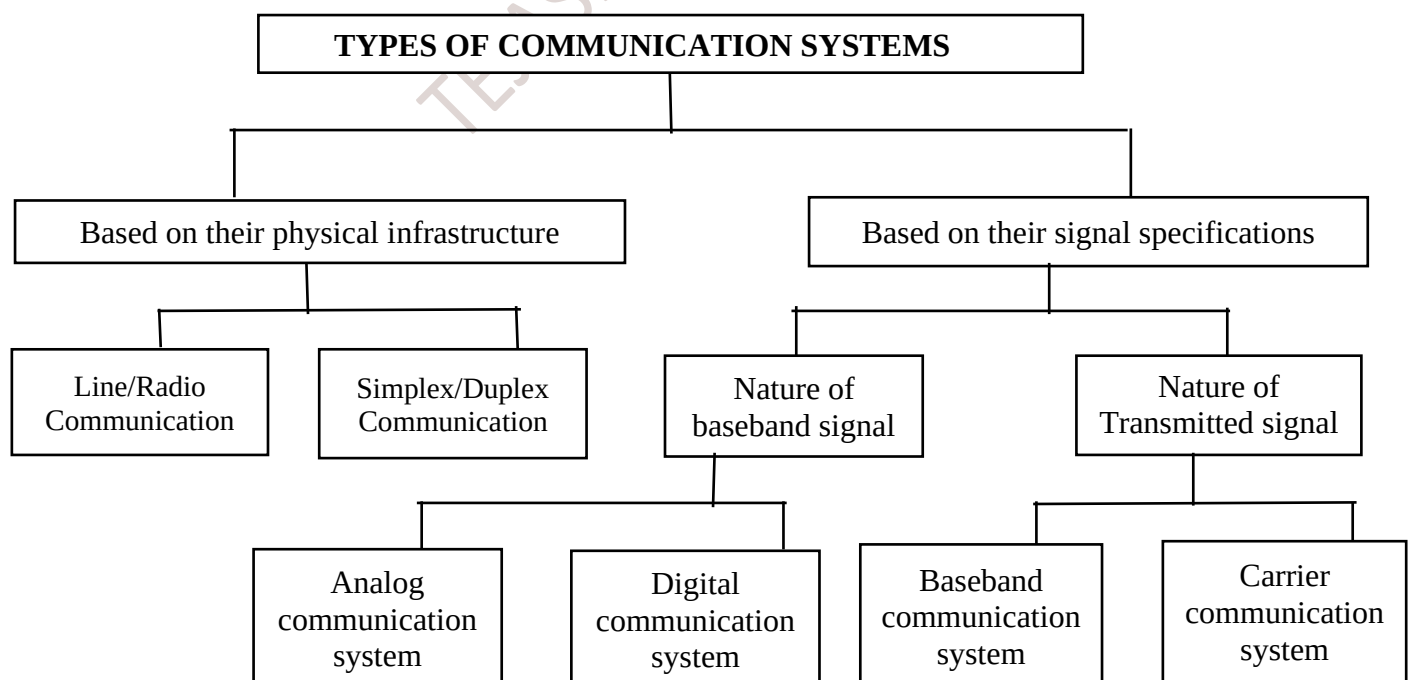


Fig 4.7: Types of Communication Systems

MODULATION

Modulation is the process of changing the parameters of the carrier signal, in accordance with the instantaneous values of the modulating signal.

TYPES OF MODULATION:

- Continuous-wave Modulation
 - Amplitude Modulation
 - Frequency Modulation
 - Phase Modulation
- Pulse modulation
 - PAM
 - PWM
 - PPM
 - PCM

1. Amplitude modulation (AM)

- AM is defined as the modulation technique in which the instantaneous amplitude of the carrier signal is varied in accordance with the instantaneous amplitude of the analog modulating signal to be transmitted while the frequency and the phase of the carrier signal remain unchanged.
- Figure 4.8 shows the high frequency carrier signal, modulating signal and the modulated signal.

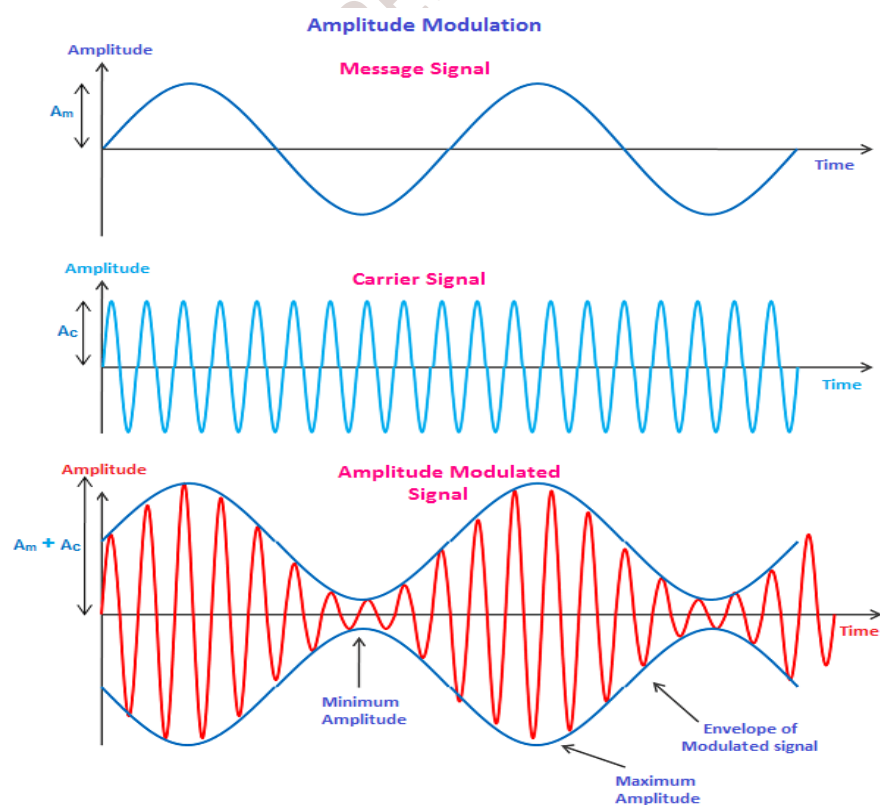


Fig: 4.8 Waveforms of Amplitude modulation

- It can be clearly seen from the figure 4.8 that the modulating signal seems to be superimposed on the carrier signal. The amplitude variations in the peak values of the carrier signal exactly replicate the modulating signal at different points in time which is known as an envelope.

2. Frequency Modulation:

- A modulating signal may vary the frequency of the carrier keeping the amplitude and phase constant. This type of modulation is called Frequency modulation. Broadly speaking, the frequency modulation is the process of changing the frequency of the carrier voltage in accordance with the instantaneous value of the modulating voltage.
- The original frequency of the carrier signal is called Centre or resting frequency and denoted by f_c . The amount by which the frequency of the carrier wave changes or shifts above or below the resting frequency is termed as frequency deviation (Δf). This means $\Delta f \propto m(t)$.
- The total variation is frequency of F.M. wave from the lowest to the highest is termed as carrier swing (CS), i.e., $US = 2 \times \text{frequency deviation in Centre frequency}$ or $CS = 2 \Delta f$.
- Modulation index in F.M. is the ratio of frequency deviation to the modulating frequency,

$$\text{i.e. } \mu_f = \frac{\text{Frequency Deviation}}{\text{Modulating Frequency}} = \frac{\Delta f}{f_m}$$

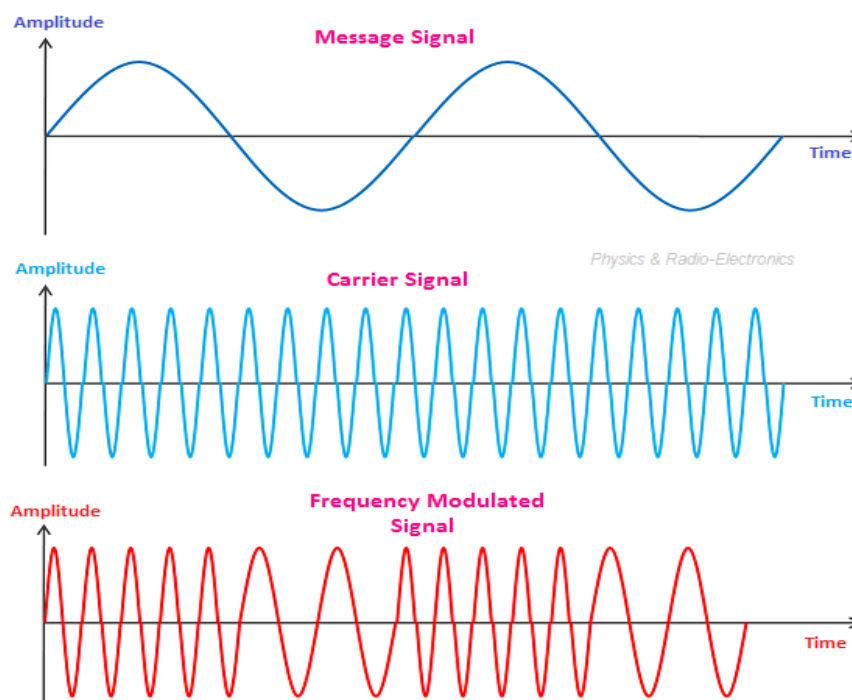


Fig: 4.9 Waveforms of Frequency modulation

3. Phase modulation:

- PM is another form of angle modulation. Phase modulation is the process in which the instantaneous phase of the carrier signal is varied in accordance with the instantaneous

amplitude of the modulating signal.

- In this type of modulation, the amplitude and frequency of the carrier signal remains unaltered after pulse modulation.

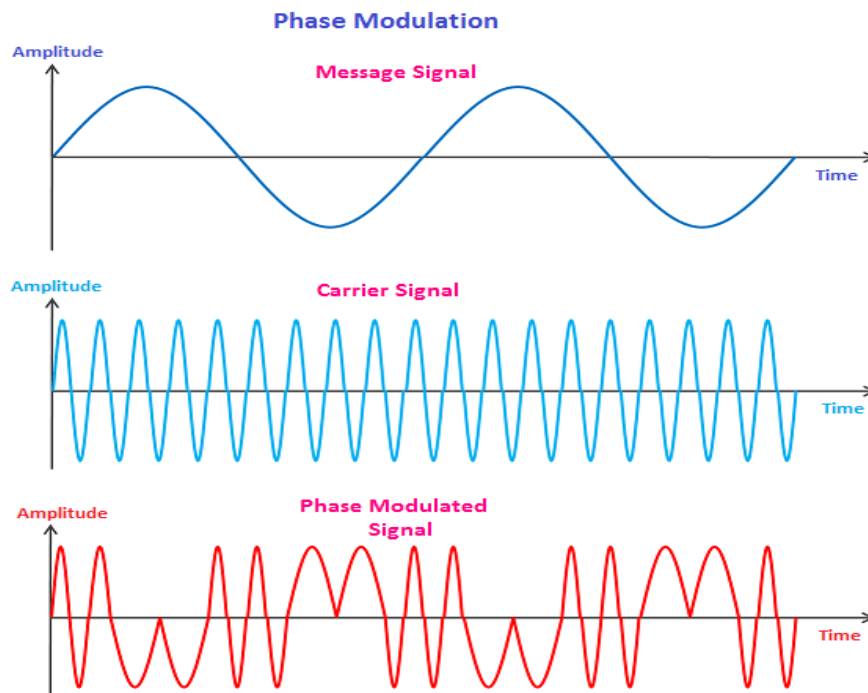


Fig: 4.10 Waveforms of Phase modulation

Pulse modulation:

- It may be used to transmit analog information, such as continuous speech or data. It is system in which continuous waveforms are sampled at regular intervals. Pulse modulation may be subdivided into two categories, analog and digital.
- Pulse-amplitude and pulse-time modulation are both analog, while the pulse code and delta modulation system are both digital.

4.1 Phase-amplitude modulation (PAM):

- PAM is the simplest form of pulse modulation. PAM is a pulse modulation system in which the signal is sampled at regular intervals, and each sample is made proportional to the amplitude of the signal at the instant of sampling. The pulses are then sent by either wire or cable, or else are used to modulate a carrier.
- The ability to use constant-amplitude pulses is a major advantage of pulse modulation, and since PAM does not use constant-amplitude pulses, it is infrequently used. When it is used, the pulses frequency-modulate the carrier.

4.2 Pulse width or pulse-duration modulation (PWM or PDM):

- In this system, the starting time and amplitude of each pulse are constant but the width or duration of each pulse is made proportional to the instantaneous value of analog signal.

- PDM has the disadvantage, when compared with pulse-position modulation (PPM), that its pulses are of varying width and therefore of varying power content. This means that the transmitter must be powerful enough to handle the maximum-width pulses, although the average power transmitted is perhaps only half of the peak power. PWM still works if synchronization between transmitter and receiver fails, whereas PPM does not.

4.3 Pulse position-modulation (PPM):

In this system, the amplitude and width of the pulses is kept constant, while the position of each pulse, in relation to the position of a recurrent reference pulse is varied by instantaneous sampled value of the modulating wave. As compared to PWM, PPM has the advantage of requiring constant transmitter power output, but the disadvantage of depending on transmitter receiver synchronization.

Pulse-code modulation (PCM):

PCM is a digital process in which the message signal is sampled is rounded off to the nearest value of a finite set of allowable values and rounded values are coded. PCM generator produces a series of numbers or digits. Each one of these digits, almost always in binary code, represents the approximate amplitude of the signal sample at that instant.

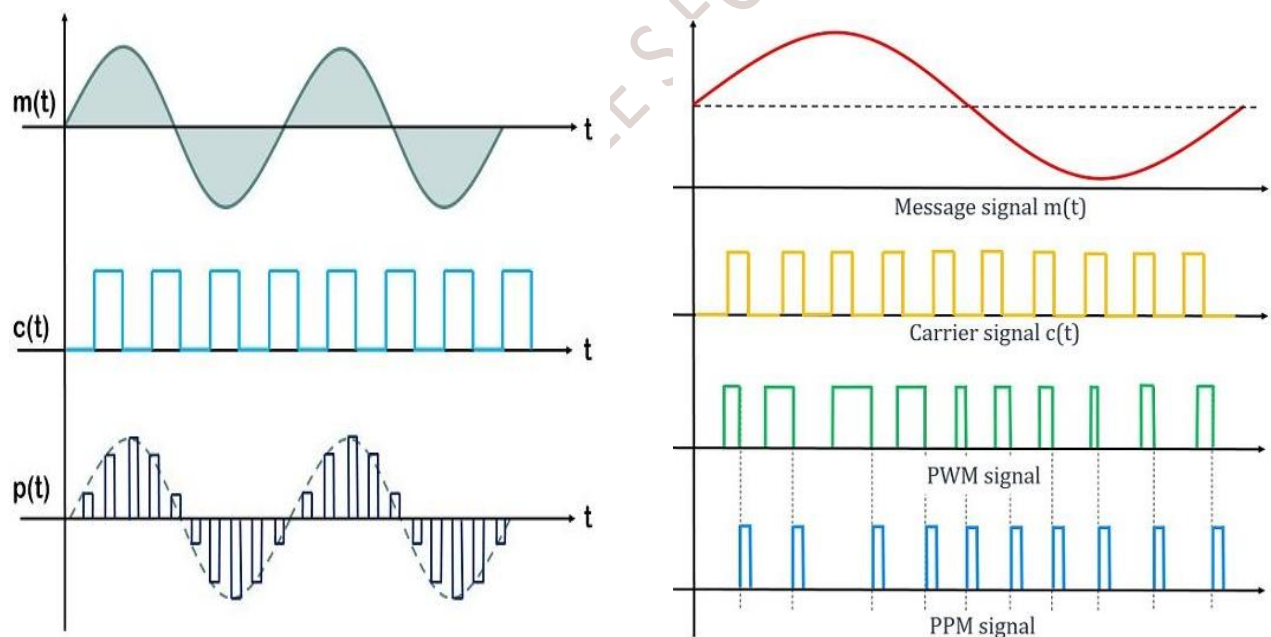


Fig 4.11: Waveforms of PAM, PWM & PPM

Radio wave propagation (Ground, space, sky):

- In *space communication* electromagnetic waves of different frequencies are used to carry information through the physical space acting as the transmission medium. Electromagnetic waves with frequencies extending from about 10 kHz to 300 GHz are classed as **radio waves**.

- Depending primarily on the frequency a radio wave travels from the transmitting to the receiving antenna in several ways. On the basis of the mode of propagation, radio waves can be broadly classified as:

(a) ground or surface wave. (b) space or tropospheric wave. (c) sky wave.

(a). Ground wave propagation:

- In ground wave propagation, radio waves are guided by the earth and move along its curved surface from the transmitter to the receiver.
- As the waves move over the ground, they are strongly influenced by the electrical properties of the ground. As high frequency waves are strongly absorbed by ground; ground wave propagation is useful only at low frequencies.
- Below 500 kHz, ground waves can be used for communication within distances of about 1500 km from the transmitter.
- AM radio broadcast in the medium frequency band cover local areas and take place primarily by the ground wave. Ground wave transmission is very reliable whatever the atmospheric conditions be.

(b). Space or tropospheric wave propagation:

- When a radio wave transmitted from an antenna, travelling in a straight line directly reaches the receiving antenna, it is termed as space or tropospheric wave.
- In space wave or line of sight propagation, radio waves move in the earth's troposphere within about 15 km over the surface of the earth.
- The space wave is made up of two components:
 - (a) a direct or line-of-sight wave from the transmitting to the receiving antenna.
 - (b) the ground-reflected wave traversing from the transmitting antenna to ground and reflected to the receiving antenna.
- Television frequencies in the range 100-220 MHz are transmitted through this mode.

(c). Sky wave propagation:

- In this mode of propagation, radio waves transmitted from the transmitting antenna reach the receiving antenna after reflection from the ionosphere, i.e., the ionized layers lying in the earth's upper atmosphere.
- Short wave transmission around the globe is possible through sky wave via successive reflections at the ionosphere and the earth's surface.

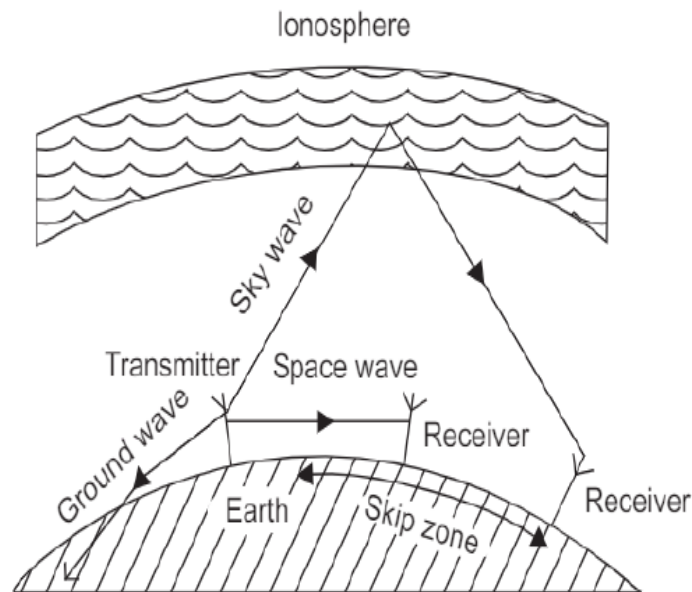


Fig 4.12: Mode of Propagation

Sampling theorem:

- There are two types of signals, continuous-time and discrete-time signals. The processing of discrete-time signals is more flexible and is also preferable than the continuous-time signals.
- The sampling theorem governs the conversion of continuous-time signal into discrete-time signal.
- The concept of sampling provides a widely used method for using discrete-time system technology to implement continuous-time systems and process the continuous-time signals.
- A continuous-time signal may be completely represented in its samples and recovered back if the sampling frequency is $f_s \geq 2f_m$. Here, f_s is the sampling frequency and f_m is the maximum frequency present in the signal.

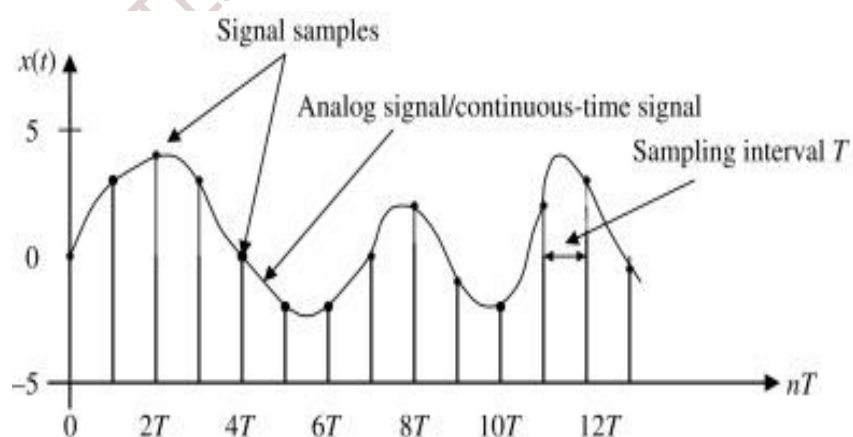


Fig 4.13: Sampling Waveform

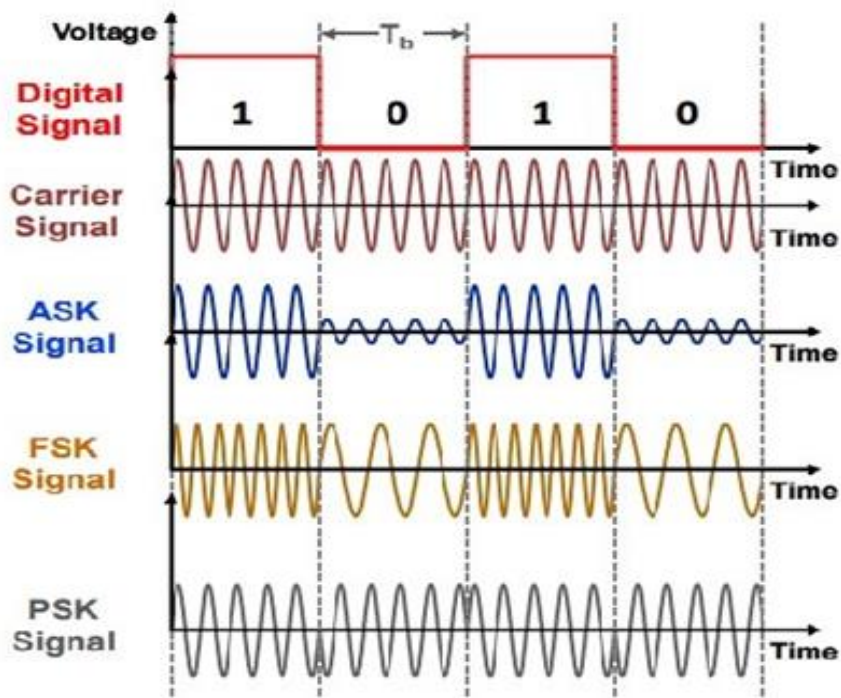
Nyquist Rate:

- When the sampling rate becomes exactly equal to $2f_m$ samples per second, then it is called Nyquist rate. Nyquist rate is the minimum sampling rate.

- A low pass filter is used to recover the original signal from its samples.
- The process of reconstructing the continuous-time signal from its samples is known as interpolation.
- When the sampling frequency is less than the Nyquist rate, aliasing problem is said to occur.
- Aliasing is the phenomenon in which a high frequency component in the frequency spectrum of the signal takes the identity of a lower frequency component in the spectrum of the sampled signal.
- To avoid aliasing:
 - Pre-alias filter must be used to limit the band of frequencies of the signal to f_m Hz.
 - Sampling frequency must be selected such that $f_s \geq 2f_m$.

Digital Modulation Schemes:

- In digital communications, the modulating signal consists of binary data. When it is required to transmit digital signals on a bandpass channel, the amplitude, frequency or phase of the sinusoidal carrier is varied in accordance with the incoming digital data.
- Since, the digital data is in discrete steps, the modulation of the bandpass sinusoidal carrier is also done in discrete steps. Due to this reason, this type of modulation is known as **digital modulation**.
- Digital modulation schemes are classified as under:
 - Amplitude Shift Keying (ASK)
 - Frequency Shift Keying (FSK)
 - Phase Shift Keying (PSK)
- Because of constant amplitude of FSK or PSK, the effect of non-linearities, noise interference is minimum on signal detection. However, these effects are more pronounced on ASK. Therefore, FSK and PSK are preferred over ASK.
- Coherent digital modulation techniques are those techniques which employ coherent detection. In coherent detection, the local carrier generated at the receiver is phase locked with the carrier at the transmitter. Thus, the detection is done by correlating the received noisy signal and locally generated carrier. The coherent detection is also called synchronous detection.
- ASK signal may be generated by simply applying the incoming binary data and the sinusoidal carrier to the two inputs of a product modulator.
- The demodulation of binary ASK waveform can be achieved with the help of coherent detector.



Radio Signal Transmission

- Fig 4.14 shows the architecture of a wireless communication transmitter. In the figure, the transmitter usually processes the information in two stages. In the first stage, a modulator accepts the incoming bits, and computes symbols that represent the amplitude and phase of the outgoing wave. It then passes these to the analogue transmitter, which generates the radio wave itself.
- The modulation scheme used in Fig. 4.14 is known as quadrature phase shift keying (QPSK).
- A QPSK modulator takes the incoming bits two at a time and transmits them using a radio wave that can have four different states. These have phases of 45° , 135° , 225° and 315° .

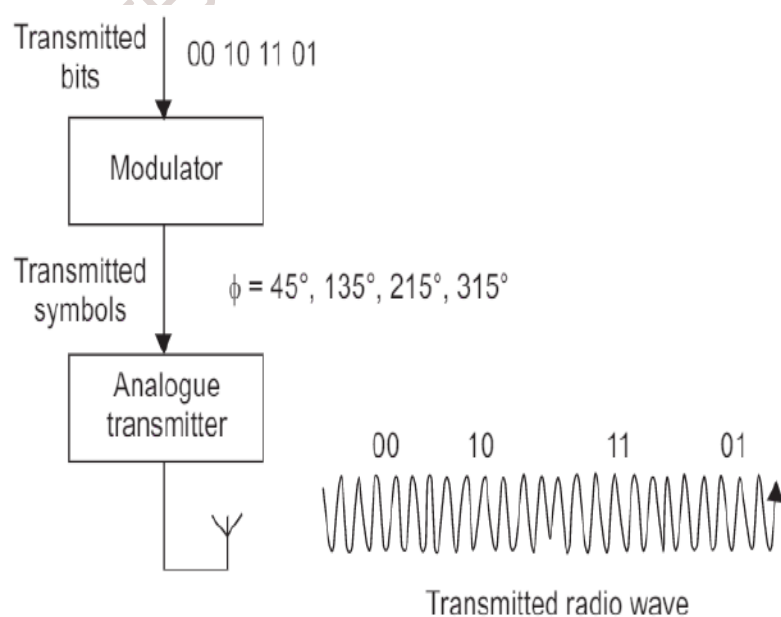
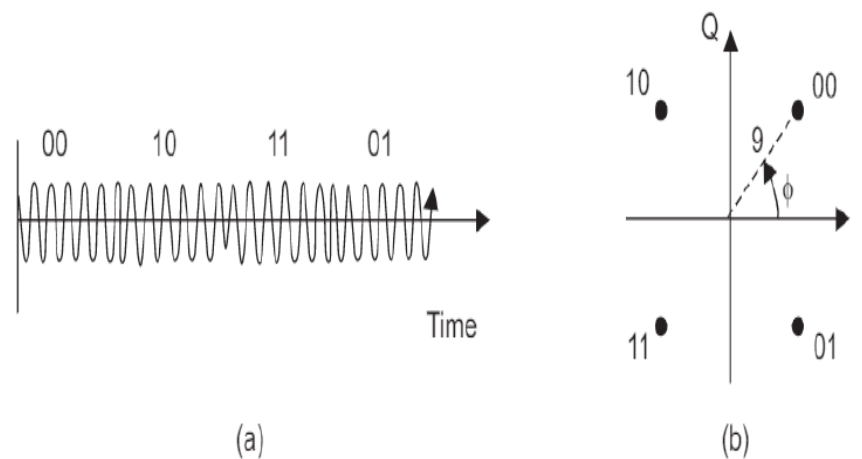


Fig 4.14: Architecture of a wireless communication transmitter



4.15: Quadrature phase shift keying (a) Example QPSK waveform (b) QPSK constellation diagram

- Fig. 4.15 (a), which correspond to bit combinations of 00, 10, 11 and 01 respectively. We can represent the four states of QPSK using the constellation diagram shown in Fig. 4.15 (b).
- In this diagram, the distance of each state from the origin represents the amplitude of the transmitted wave, while the angle (measured anti-clockwise from the x-axis) represents its phase. Usually, it is more convenient to represent each symbol using two other numbers, which are known as the in-phase (I) and quadrature (Q) components. These are computed as follows:

$$I = a \cos \phi$$

$$Q = a \sin \phi$$

where a is the amplitude of the transmitted wave and ϕ is its phase.

- Mathematicians will recognize the in-phase and quadrature components as the real and imaginary parts of a complex number.
- As shown in Fig. 4.16, LTE uses four modulation schemes altogether. Binary phase shift keying (BPSK) sends bits one at a time, using two states that can be interpreted as starting phases of 0° and 180° , or as signal amplitudes of +1 and -1.
- LTE uses this scheme for a limited number of control streams, but does not use it for normal data transmissions.
- 16 quadrature amplitude modulation (16-QAM) sends bits four at a time, using 16 states that have different amplitudes and phases.
- Similarly, 64-QAM sends bits six at a time using 64 different states, so it has a data rate six times greater than that of BPSK.

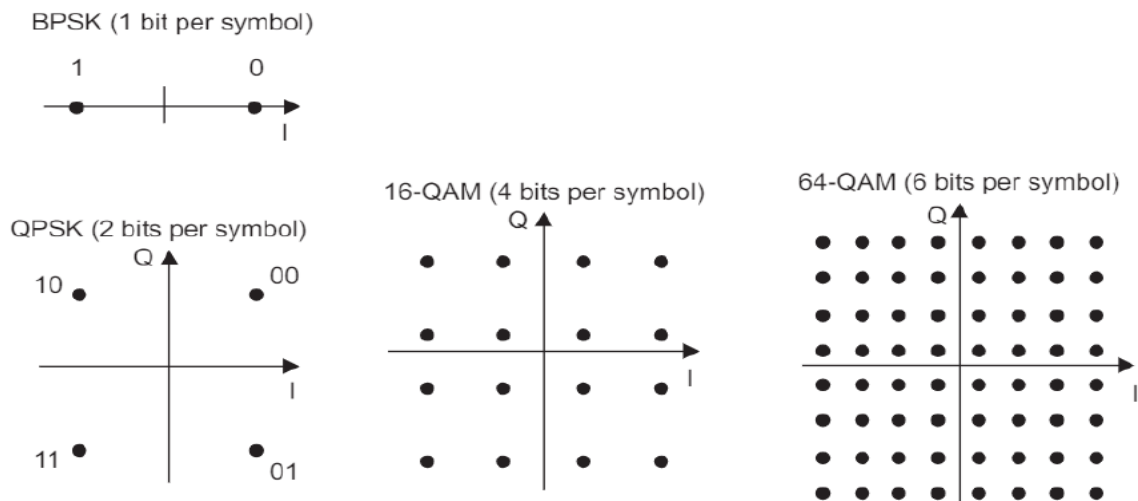


Fig 4.16: Modulation schemes used by LTE

Multiple Access Techniques:

- The techniques described so far work well for one-to-one communications. In a cellular network, however, a base station has to transmit to many different mobiles at once. It does this by sharing the resources of the air interface, in a technique known as multiple access.
- Mobile communication systems use a new different multiple access techniques, two of which are shown in Fig. 4.17 **frequency division multiple access (FDMA)** was used by the first-generation analogue systems. In this technique, each mobile receives on its own carrier frequency, which it distinguishes from the others by the use of analogue filters.
- In **time division multiple access (TDMA)**, mobiles receive information on the same carrier frequency but at different times.
- **GSM** uses a mix of frequency and time division multiple access, in which every cell has several carrier frequencies that are each shared amongst eight different mobiles.
- LTE uses another mixed technique known as orthogonal frequency division multiple access (**OFDMA**).

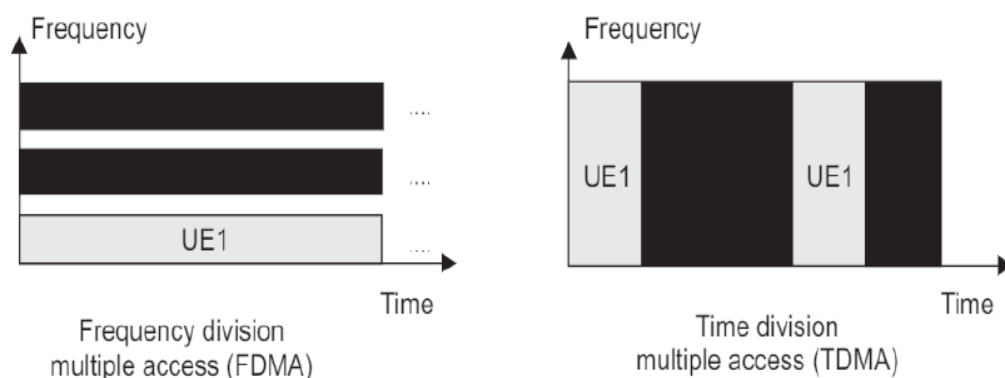


Fig 4.17: Example multiple access techniques

- Third generation communication systems used a different technique altogether, known as **code division multiple access (CDMA)**.
- In this technique, mobiles receive on the same carrier frequency and at the same time, but the signals are labelled by the use of codes, which allow a mobile to separate its own signal from those of the others.
- LTE uses a few of the concepts from CDMA for some of its control signals, but does not implement the technique otherwise. Multiple access is actually a generalization of a simpler technique known as multiplexing.

FDD and TDD Modes

- A mobile communication system can operate in the transmission modes as shown in Fig. 4.18.
- When using frequency division duplex (FDD), the base station and mobile transmit and receive at the same time, but using different carrier frequencies.
- Using time division duplex (TDD), they transmit and receive on the same carrier frequency but at different times. FDD and TDD modes have different advantages and disadvantages.
- In FDD mode, the bandwidths of the uplink and downlink are fixed and are usually the same. This makes it suitable for voice communications, in which the uplink and downlink data rates are very similar.
- In TDD mode, the system can adjust how much time is allocated to the uplink and downlink. This makes it suitable for applications such as web browsing.
- TDD mode can be badly affected by interference if, for example, one base station is transmitting while a nearby base station is receiving.
- To avoid this, nearby base stations must be carefully time synchronized and must use the same allocations for the uplink and downlink, so that they all transmit and receive at the same time.
- This makes TDD suitable for networks that are made from isolated hotspots, because each hotspot can have a different timing and resource allocation. In contrast, FDD is often preferred for wide-area networks that have no isolated regions.

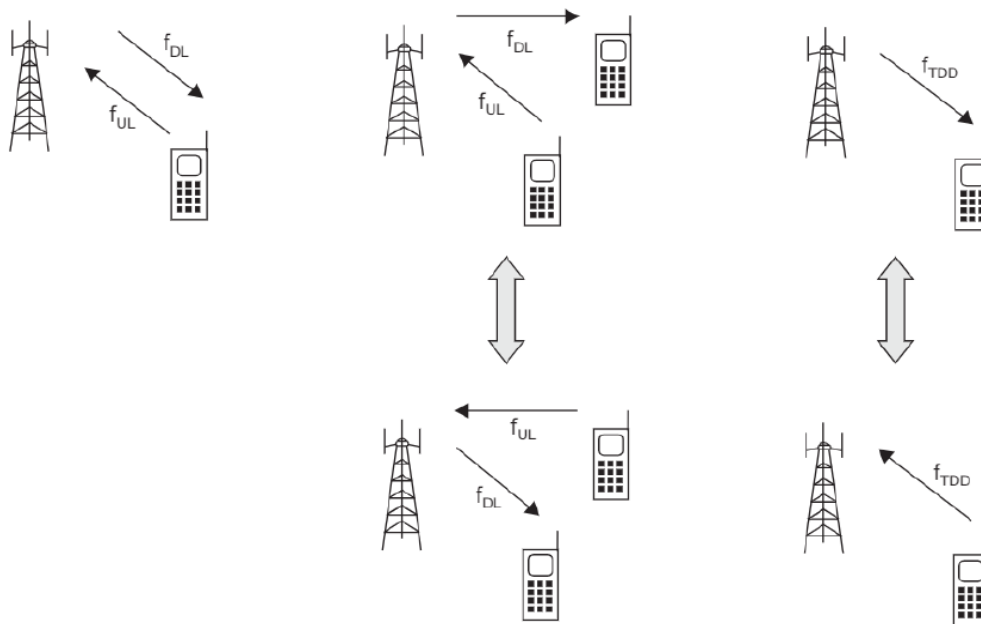


Fig 4.18: Operation of FDD and TDD modes

- When operating in FDD mode, the mobile usually has to contain a high attenuation duplex filter that isolates the uplink transmitter from the downlink receiver.
- LTE supports each of the modes described above. A cell can use either FDD or TDD mode.

MULTIPATH AND FADING

- Propagation loss and noise are not the only problem. As a result of reflections, rays can take several different paths from the transmitter to the receiver. This phenomenon is known as **multipath**.
- At the receiver, the incoming rays can add together in different ways, which are shown in Fig. 4.19. If the peaks of the incoming rays coincide then they reinforce each other, a situation known as **constructive interference**.
- If, however, the peaks of one ray coincide with the troughs of another, then the result is destructive interference, in which the rays cancel.
- **Destructive interference** can make the received signal power drop to a very low level, a situation known as **fading**. The resulting increase in the error rate makes fading a serious problem for any mobile communication system.

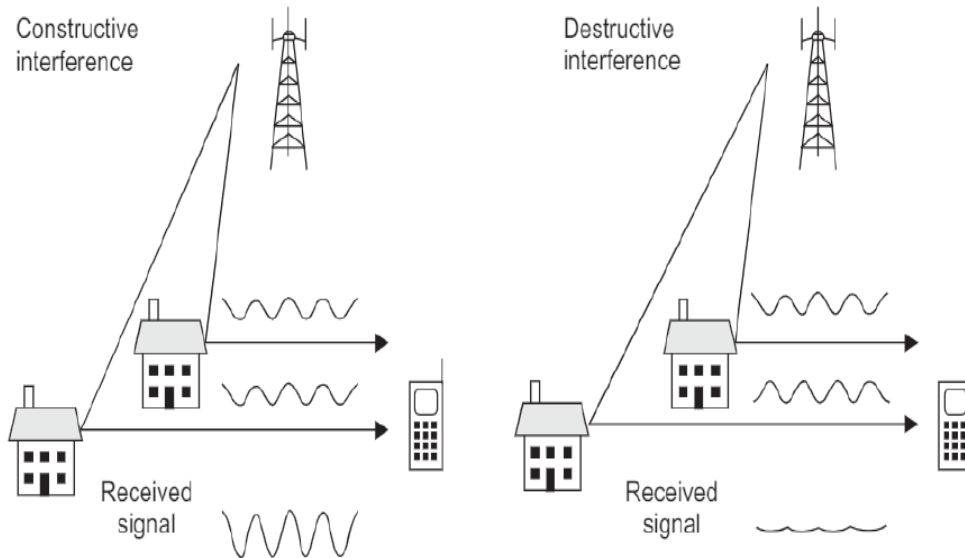


Fig 4.19: Generation of constructive interference, destructive interference and fading in a multipath environment

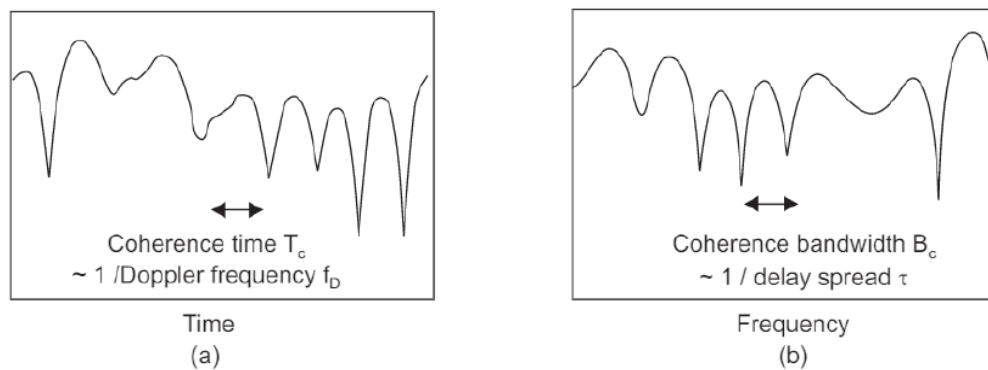


Fig 4.20: Fading as a function of (a) time and (b) frequency

- If the mobile moves from one place to another, then the ray geometry changes, so the interference pattern changes between constructive and destructive.
- Fading is therefore a function of time, as shown in Fig. 4.20 (a). The amplitude and phase of the received signal vary over a timescale called the *coherence time*, T_c which can be estimated as follows:

$$T_c \approx \frac{1}{f_D}$$

Here f_D is the mobile's Doppler frequency:

$$f_D = \frac{v}{c} f_c$$

where f_c is the carrier frequency, v is the speed of the mobile and c is the speed of light (3×10^8 m/s).

- For example, a pedestrian might walk with a speed of 1 ms^{-1} (3.6 km hr^{-1}). At a carrier frequency of 1500 MHz, the resulting Doppler shift is 5 Hz, giving a coherence time of about 200

milliseconds. Faster mobiles move through the interference pattern more quickly, so their coherence time is correspondingly less.

- If the carrier frequency changes, then the wavelength of the radio signal changes. This also makes the interference pattern change between constructive and destructive, so fading is a function of frequency as well 4.20 (b).

Error Management

Forward Error Correction:

- We know that noise and interference lead to errors in a wireless communication receiver. These are bad enough during voice calls, but are even more damaging to important information such as web pages and emails. Fortunately, there are several ways to solve the problem. The most important technique is **forward error correction**.
- In this technique, the transmitted information is represented using a *codeword* that is typically two or three times as long. The extra bits supply additional, redundant data that allow the receiver to recover the original information sequence.
- For example, a transmitter might represent the information sequence 101 using the codeword 110010111. After an error in the second bit, the receiver might recover the codeword 100010111.
- If the coding scheme has been well designed, then the receiver can conclude that this is not a valid codeword, and that the most likely transmitted codeword was 110010111.
- The receiver has therefore corrected the bit error and can recover the original information. The effect is very like written English, which contains redundant letters that allow the reader to understand the underlying information, even in the presence of spelling mistakes.
- The coding rate is the number of information bits divided by the number of transmitted bits (1/3 in the example above). Usually, forward error correction algorithms operate with a fixed coding rate.
- Despite this, a wireless transmitter can still adjust the coding rate using the two- stage process shown in Fig. 4.21.
- In the first stage, the information bits are passed through a fixed-rate coder. The main algorithm used by LTE is known as Turbo coding and has a fixed coding rate of 1/3.
- In the second stage, called rate matching, some of the coded bits are selected for transmission, while the others are discarded in a process known as puncturing.
- The receiver has a copy of the puncturing algorithm, so it can insert dummy bits at the points where information was discarded. It can then pass the result through a turbo decoder for error correction.

- Changes in the coding rate have a similar effect to changes in the modulation scheme. If the coding rate is low, then the transmitted data contain many redundant bits.
- This allows the receiver to correct a large number of errors and to operate successfully at a low SINR, but at the expense of a low information rate.
- If the coding rate is close to 1, then the information rate is higher but the system is more vulnerable to errors.
- LTE exploits this with a similar trade-off to the one we saw earlier, by transmitting with a high coding rate if the received SINR is high and vice versa.

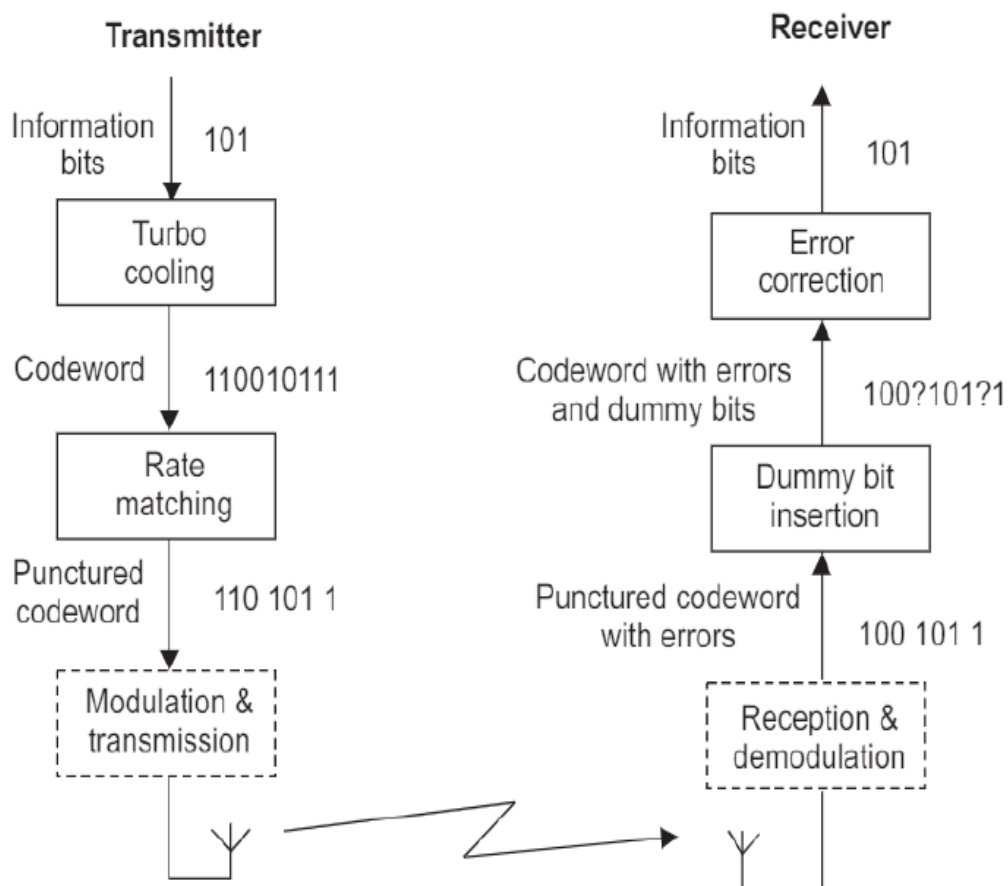


Fig 4.21: Block diagram of a transmitter and receiver using forward error correction and rate Matching

Automatic Repeat Request:

- Automatic repeat request (ARQ) is another error management technique, which is illustrated in Fig 4.22. Here, the transmitter takes a block of information bits and uses them to compute some extra bits that are known as a cyclic redundancy check (CRC).
- It appends these to the information block and then transmits the two sets of data in the usual way.

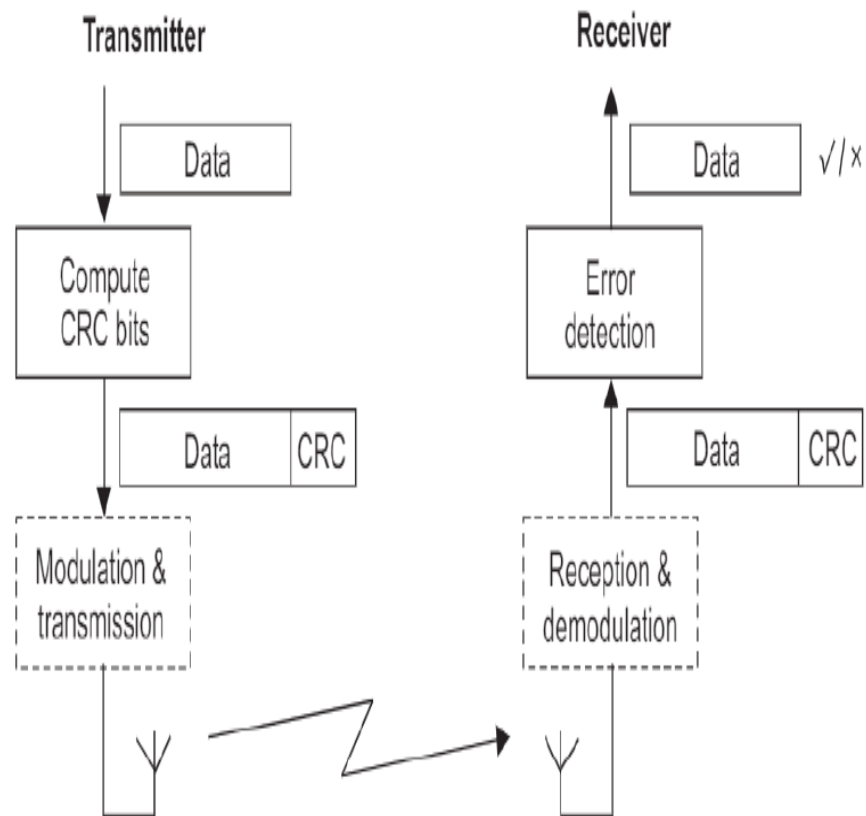


Fig 4.22: Block diagram of a transmitter and receiver using automatic repeat request

- The receiver separates the two fields and uses the information bits to compute the expected CRC bits. If the observed and expected CRC bits are the same, then it concludes that the information has been received correctly and sends a positive acknowledgement back to the transmitter.
- If the CRC bits are different, it concludes that an error has occurred and sends a negative acknowledgement to request a re-transmission.
- Positive and negative acknowledgements are often abbreviated to ACK and NACK respectively.
- A wireless communication system often combines the two error management techniques that we have been describing. Such a system corrects most of the bit errors by the use of forward error correction and then uses automatic repeat requests to handle the remaining errors that leak through.
- Normally, ARQ uses a technique called selective re-transmission Fig 4.23 in which the receiver waits for several blocks of data to arrive before acknowledging them all.
- This allows the transmitter to continue sending data without waiting for an acknowledgement, but it means that any re-transmitted data can take a long time to arrive.
- Consequently, this technique is only suitable for non-real-time streams such as web-pages and emails.

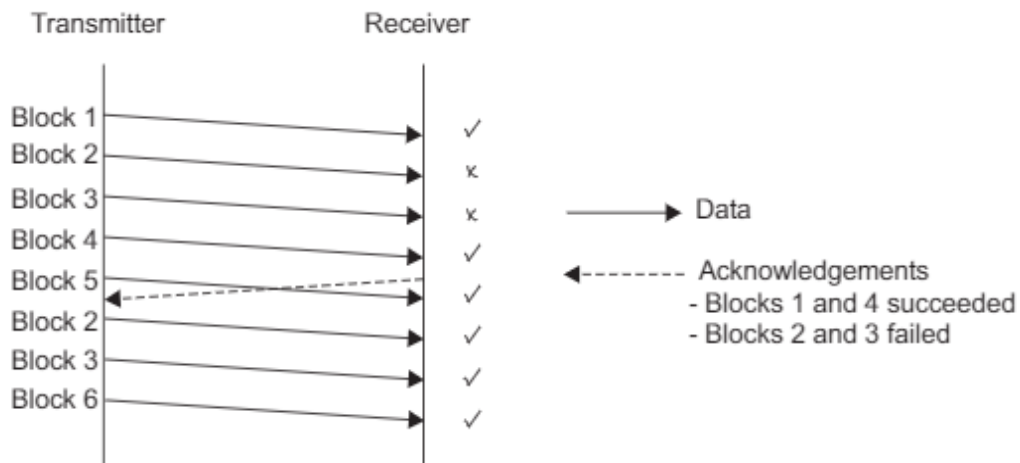


Fig 4.23: Operation of a selective re-transmission ARQ scheme

Antenna:

- An antenna is a device for converting electromagnetic radiation in space into electrical currents in conductors or vice-versa, depending on whether it is being used for receiving or for transmitting, respectively.
- Antennas transform wire propagated waves into space propagated waves. They receive electromagnetic waves and pass them onto a receiver or they transmit electromagnetic waves which have been produced by a transmitter.
- The requirements on the antennas needed for the ever-expanding networks are becoming continually

higher. An antenna must have the following features:

- Strictly defined radiation patterns for a most accurate network planning.
- Growing concern for the level of intermodulation due to the radiation of many HF- carriers via one antenna.
- Dual polarization.
- Electrical down-tilting of the vertical diagram.
- Unobtrusive design.

Types of antennas:**1. Omnidirectional Antenna**

An omnidirectional antenna is an antenna that has a non-directional pattern (circular pattern) in a given plane with a directional pattern in any orthogonal plane. Examples of omnidirectional antennas are *dipoles* and *collinear antennas*.

2. Dipole Antennas

A dipole antenna most commonly refers to a half-wavelength ($L/2$) dipole. The physical antenna is constructed of conductive elements whose combined length is about half of a wavelength at its

intended frequency of operation. This is a simple antenna that radiates its energy out toward the horizon.



Fig 4.24: Dipole antenna

3. Collinear Omni Antennas:

In order to create an omnidirectional antenna with higher gain, multiple omni directional structures (either wires or elements on a circuit board) can be arranged in a vertical, linear fashion to retain the same omnidirectional pattern in the azimuth plane but a more focused elevation plane beam which then has higher gain. This is frequently referred to as a collinear array.

4. Directional Antennas

- A directional antenna is one that radiates its energy more effectively in one (or some) direction than others. Typically, these antennas have one main lobe and several minor lobes. Examples of directional antennas are *patches* and *dishes*.
- Directional antennas are used for coverage as well as point-to-point links. They can be patch antennas, dishes, horns or a whole host of other varieties. They all accomplish the same goal: radiating their energy out in a particular direction.

5. Patch Antennas

- A patch antenna, in its simplest form, is just a single rectangular (or circular) conductive plate that is spaced above a ground plane.
- Patch antennas are attractive due to their low profile and ease of fabrication. The radiation pattern of a single patch is characterized by a single main lobe of moderate beamwidth.
- Frequently, the beamwidths in the azimuth and elevation planes are similar, resulting in a fairly circular beam, although this is by no means universal.
- The beamwidths can be manipulated to produce an antenna with higher or lower gain, depending on the requirements. An antenna built with a single patch will have a maximum gain of about 9 dBi or a bit less.
- The patch antenna in Fig. 4.25 shows how simple these antennas can be. This is a simple rectangular patch built over a rectangular ground plane.

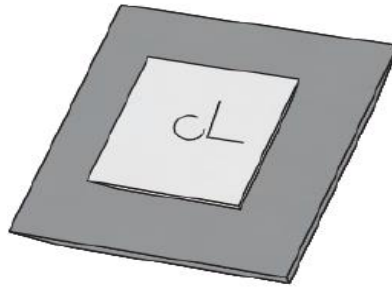


Fig 4.25: Patch antenna

6. Patch Array Antennas:

- A patch array antenna is, in general, some arrangement of multiple patch antennas that are all driven by the same source. Frequently, this arrangement consists of patches arranged in orderly rows and columns (a rectangular array) as shown in Fig. 4.26.
- The reason for these types of arrangements is higher gain. Higher gain commonly implies a narrower beamwidth and that is, indeed, the case with patch arrays. The array shown here has a gain of about 18 dBi with an azimuth and elevation plane beamwidth of about 20 degrees.

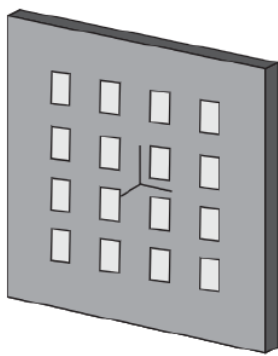


Fig 4.26: 4x4 patch array antenna

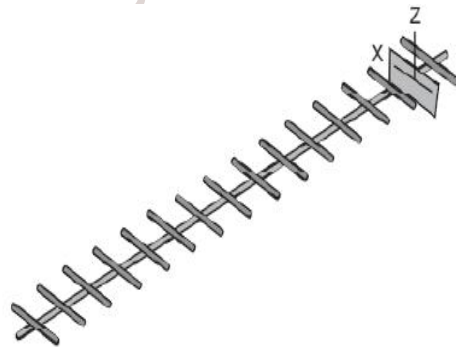


Fig 4.27: Yagi-Antenna model

7. Yagi Antennas

- A Yagi antenna is formed by driving a simple antenna, typically a dipole or dipole-like antenna, and shaping the beam using a well-chosen series of non-driven elements whose length and spacing are tightly controlled.
- The Yagi shown here in Fig. 4.27 is built with one reflector (the bar behind the driven antenna) and 14 directors (the bars in front of the driven antenna).
- Many times, these antennas are designed so that they can be rotated for either horizontal or vertical polarization, so having the same 3-dB beamwidth in each plane is a nice feature in those instances. Again, the Yagi antenna is a directional antenna that radiates its energy out in one main direction. Their directional nature seems to be somewhat intuitive due to their common, tubular form factor. It is easy to visualize aiming these antennas much like a rifle.